

Roll Number: _____

Date: 5 Dec 2024, Time: 2:00 PM



Thapar Institute of Engineering and Technology, Patiala

Department of Computer Science and Engineering

END-Semester Examination

BE 3rd Sem | UTA018: Object Oriented Programming | Time: 3 Hours | Max Marks:40

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Instructions: 1. Attempt any five questions.

2. If you attempt all the questions, first five questions will be considered for evaluation.

3. Assume all the necessary libraries are included in all the codes of this paper.

Q.N.	Questions	Marks	CO	BL
Q.1	a) Explain the role of public and private access specifiers in C++ classes with suitable programming example.	4	CO1	L2
	b) Write a C++ program to demonstrate constructor overloading within a single class, by including default, parameterized, and copy constructor.	4	CO1	L3
Q.2	a) Explain three components of STL.	3	CO4	L2
	b) Write a C++ program to store integers in a vector and perform the following operations on the vector . I. Add a new integer at the end. II. Remove the integer from the end III. Sort the vector in ascending order. IV. Display the final vector after all operations.	5	CO4	L3
Q.3	a) Implement a class for the main() function given below: <pre>int main () { Demo d1(10); Demo d2(20); if(d1==d2) cout<<"both are equal"<<endl; else cout<<"not equal"<<endl; if(d1<d2) cout<<"d1 is less than d2"<<endl; else cout<<"d2 is less than d1"<<endl; }</pre>	4	CO3	L4
	b) Implement a class in C++ to perform class-to-class conversion that converts energy from Joules to Calories. (i) Create a Calorie class with a private data member to store energy in Calories with default and parameterized constructors to initialize. (ii) Create a Joule class with a private data member to store energy in Joules with default and parameterized constructors to initialize. Use the type casting operator function that converts energy from Joules to Calories (1 Joule = 0.239006 Calories). The program should take energy in Joules as input, convert it to Calories, and display the result. Ex: Input: 1000. Output: Energy in Calories: 239.006. The main () function for the same is given below: <pre>int main() { float energyInJoules; Calorie calorieObj; cout << "Enter energy in Joules: "; cin >> energyInJoules; Joule jouleObj(energyInJoules); calorieObj = jouleObj; cout << "Energy in Calories: " << calorieObj.getCalories() << endl; }</pre>	4	CO3	L3
Q.4	a) Explain the concept of non-type function template parameters with the small code snippet.	2	CO4	L3
	b) Design a templated class team , that represents a Team (for a College Sports System) , with information about its players. The goal is to allow flexibility in how teams are represented—either by player names (strings) or jersey numbers (integers). The Team class contains the following data members; T players[20] (An array of players (templated as T) with a maximum size of 20), playerCount (An integer to track the current number of players in the team), int matchesPlayed (An integer for the total number of matches played), and int matchesWon (An integer for the number of matches won). Below partial is the code of the Team class template. Complete the code by performing the following task. I. Implement the default and parameterized constructors. Ensure the parameterized constructor properly initializes the players array.	6	CO4	L4

[P.T.O]

	II. Implement the addPlayer() and removePlayer() Functions: Ensure players are added only if the team has space (max 20 players). Ensure players are removed only if they exist in the team. <pre> template <typename T> class Team { private: T players[20]; int playerCount, matchesPlayed, matchesWon; public: Team() {...}; Team(T playersArray[], int count, int played, int won) {...}; void addPlayer(T player) {...}; void removePlayer(T player) {...}; }; </pre> EXAMPLE: Input: int players[] = {7, 10, 15}; Team<int> team1(players, 3, 5, 3); Output: Players: 7, 15, 20 Matches Played: 5 Matches Won: 3 Input: string players[] = {"Alice", "Bob", "Charlie"}; Team<string> team1(players, 3, 5, 3); Output: Players: Alice, Bob, Charlie Matches Played: 5 Matches Won: 3			
Q.5	Consider below given code. The data member REVERSE::x is a signed 32-bit integer and the member function REVERSE::reverse() returns x with its digits reversed. If reversing x causes the value to go outside the signed 32-bit integer range $[-2^{31}, 2^{31}-1]$, then REVERSE::reverse() throws a string ("Error") and return zero. Assume the environment does not allow you to store 64-bit integers (signed or unsigned). Give the best-fit C++ statements (if required, otherwise mention NULL) for the blank spaces (B1 to B8) in the given code below, so that the code executes successfully. NOTE: Your answer must have <u>statements for the blank spaces in the sequence of B1 to B8. NO NEED TO WRITE COMPLETE CODE.</u> <pre> class REVERSE { protected: int x; public: int reverse() { long result = 0; long copy = x > 0 ? x : -(long)x; while (copy != 0) { try { result *= 10; B1 } //try close catch(B2) { B3 } result += (copy % 10); copy /= 10; } //loop closed return x > 0 ? result : (-result); } //function end public: REVERSE(int x) { this->x=x; } virtual void reverse(REVERSE *p)=0; }; //class closed </pre> <pre> class child: public REVERSE { public: int reverse() {return 0;} public: child(int x) B4 { this->x=x; } //constructor closed void reverse(B5) { cout<< B6 ;} }; //child class closed int main() { REVERSE *p= B7 ; p-> B8 ; return 0; } //main closed </pre>	8	CO3	L4
Q.6	a) Explain Multiple and Hierarchical Inheritance with diagrams and syntax. b) Create an abstract class Employee with a pure virtual function increment_sal() and a virtual display() function. Derive two classes, ScientistCatA and ScientistCatB: I. ScientistCatA increments salary by 10%. II. ScientistCatB increments salary by 5%. In the main() function, demonstrate runtime polymorphism using an Employee* pointer, to call increment_sal() and display() for both objects.	4 4	CO2 CO2	L2 L3

Marks Distribution

Bloom's Level wise Marks Distribution



Level 2 Level 3 Level 4

Course Outcome wise Marks Distribution (EST)

